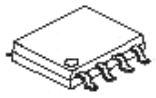
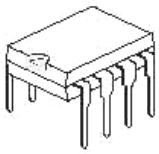
JRC4558
Dual Operational Amplifier

DESCRIPTION

The JRC4558 is a high performance monolithic dual operational amplifier.



SOP-8

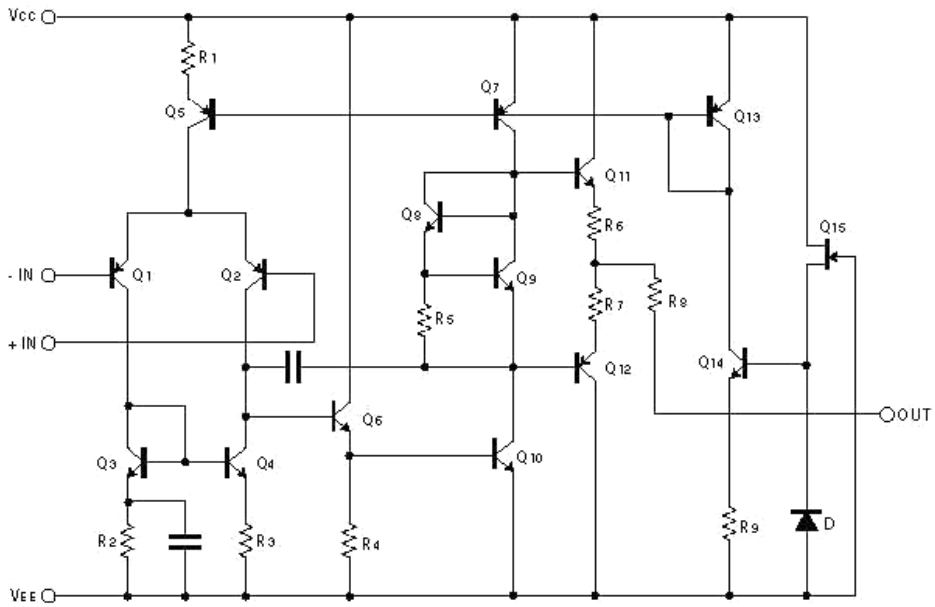


DIP-8

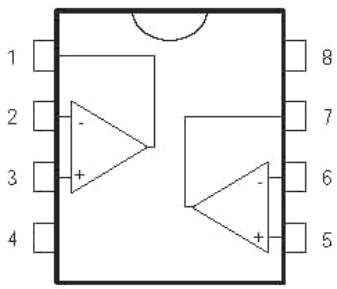
FEATURES

- No frequency compensation required
- No latch – up
- Large common mode and differential voltage range
- Parameter tracking over temperature range
- Gain and phase match between amplifiers
- Internally frequency compensated
- Low noise input transistors
- Pin to pin compatible with MC1458/LM358

BLOCK DIAGRAM (ONE SECTION ONLY)

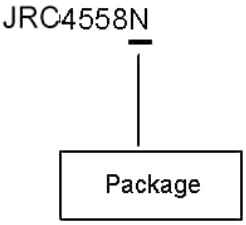


PIN CONFIGURATION



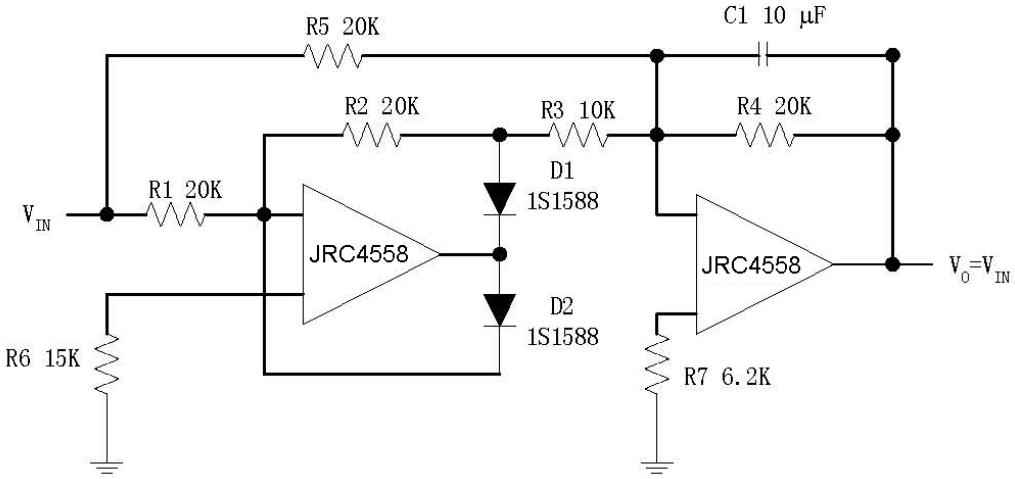
1-Output 1
2-Inverting input 1
3-Non-inverting input1
4-Vcc
5-Non-inverting input 2
6-Inverting input 2
7-Output 2
8-Vcc +

ORDERING INFOR MATION



Blank SO-8
N=PDIP8
A=SO-8 & taping

Typical Application



MAXIMUM RATINGS

PARAMETER	SYMBOL	VALUE	UNIT
Supply Voltage	Vcc	±22	V
Differential Input Voltage	VI(DIFF)	±18	V
Input Voltage	VI	±15	V
Operating Temperature	TOPR	-20~ +85	℃
Power Dissipation P-DIP 8 SOP 8	PD	600 400	mW
Storage Temperature Range	TSTG	-65~+150	℃

ELECTRICAL CHARACTERISTICS (Vcc=15.0V, VEE=-15V, TA=25℃, unless otherwise specified)

PARAMETER	SYMBOL	TEST CONDUCTION	MIN	TYP	MAX	UNIT
Supply Current, all Amp, no load	Icc			2.3	4.5	mA
Input offset voltage	V_{IO}	Rs<10KΩ		2	6	mV
Input offset current	I_{IO}			5	200	nA
Input bias current	I_{BIAS}			30	500	nA
Large signal voltage gain	GV	Vo(p-p)= ±10V, RL≤2kΩ	20	200		V/mV
Common Mode Input Votage Range	$V_{I(R)}$		±12	±13		V

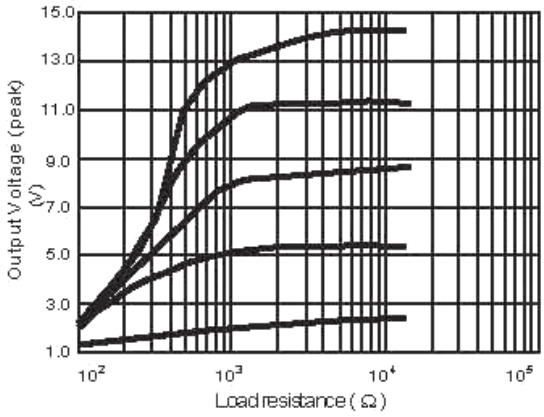
Common Mode Rejection Ratio	CMRR	Rs ≤10kΩ	70	90		dB
Supply Voltage Rejection Ratio	PSRR	Rs ≤10kΩ	76	90		dB
Output Voltage swing	Vo(p-p)	R_L≥10kΩ		±12	±14	V
Power Consumption	Pc			70	170	mV
Slew Rate	SR	Vi=±10V, RL≥2kΩ, CL≤100pF	1.2	2.2		V/μS
Rise Time	T_{RIS}	Vi=±20mV, RL≥2kΩ, CL≤100pF		0.3		μs
Overshoot	OS	Vi=±20mV, RL≥2kΩ, CL≤100pF		15		%
Input Resistance	Ri		0.3	2		MΩ
Output Resistance	Ro			75		Ω
Total Harmonic Distortion	THD	f=1KHz, Av=20dB, RL=2kΩ, Vo=2Vpp, CL=100pF		0.008		%
Channel Separation	Vo1/Vo2			120		dB

FREQUENCY CHARACTERISTICS (Ta=25°C, Vcc=15V, Vee=-15V)

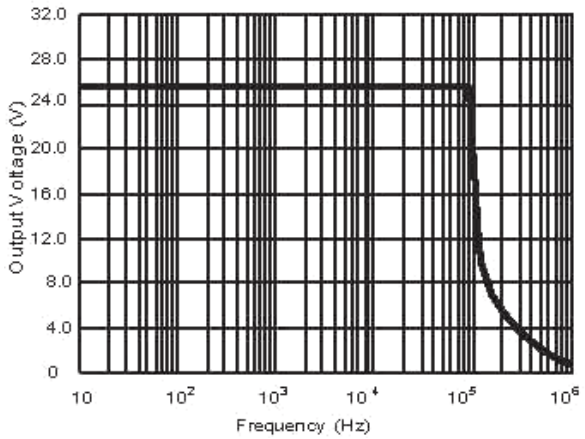
PARAMETER	SYMBOL	TEST CONDUCTION	MIN	TYP	MAX	UNIT
Unity Gain Bandwidth	BW		2.0	2.8		MHz

TYPICAL PERFORMANCE CHARACTERISTICS

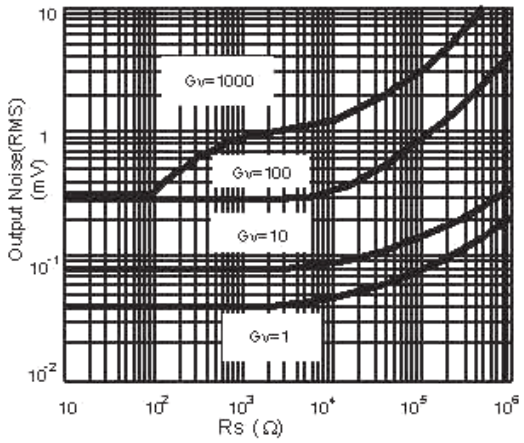
Positive output voltage swing vs load resistance



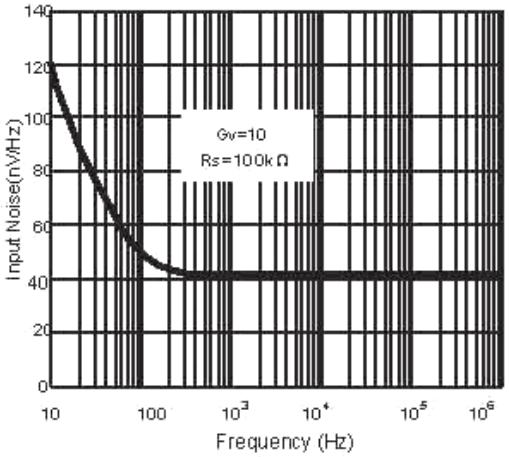
Power Bandwith (Large Signal)



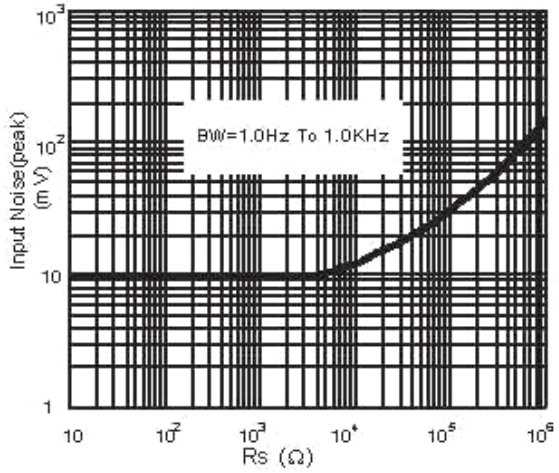
Output Noise vs Rs



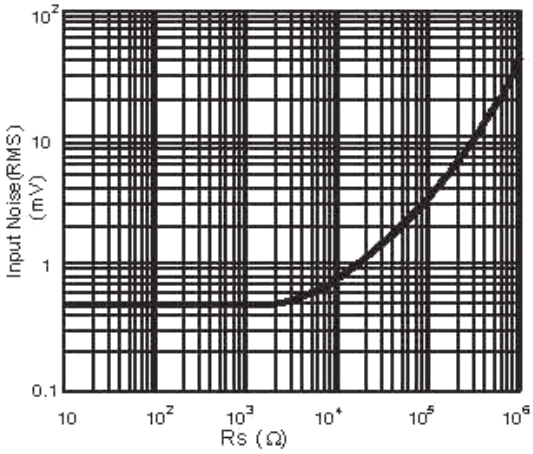
Spectral Noise Density



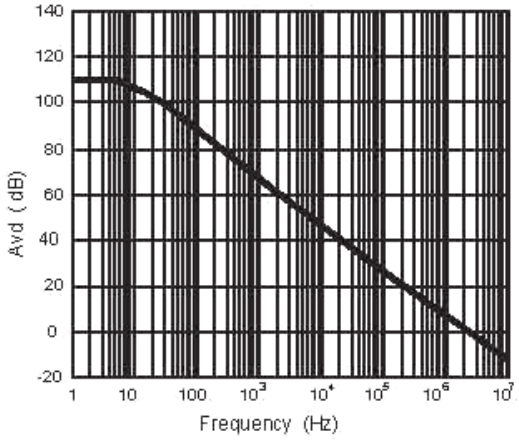
Burst Noise vs Rs



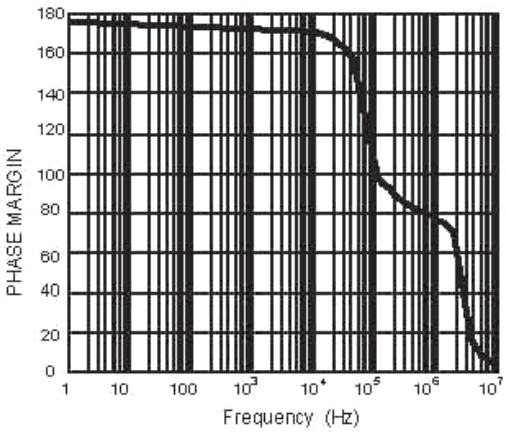
RMS Noise vs Rs



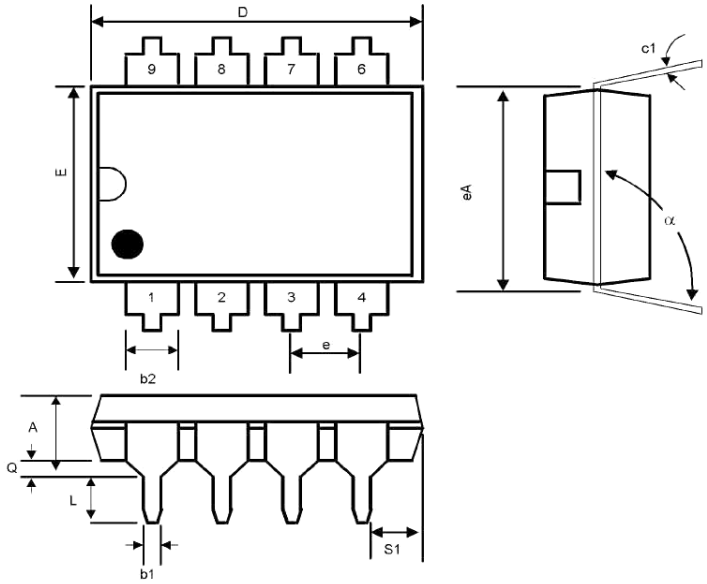
Open loop frequency response



PHASE MARGIN vs FREQUENCY

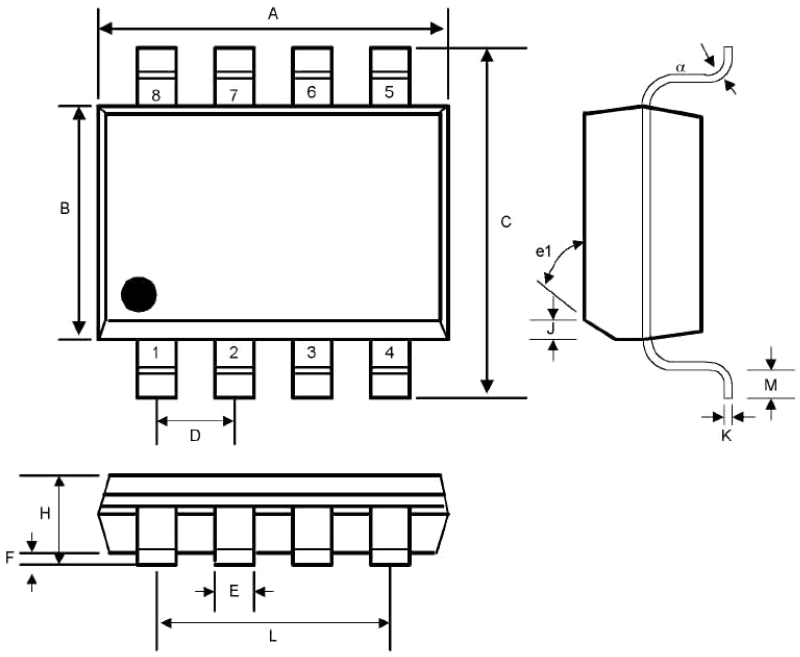


Package Outlines: DIP-8



SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	-	0.200	-	5.08	-
b1	0.014	0.023	0.36	0.58	-
b2	0.045	0.065	1.14	1.65	-
c1	0.008	0.015	0.20	0.38	-
D	0.355	0.400	9.02	10.16	-
E	0.220	0.310	5.59	7.87	-
e	0.100 BSC		2.54 BSC		-
eA	0.300 BSC		7.62 BSC		
L	0.125	0.200	3.18	5.08	-
Q	0.015	0.060	0.38	1.52	-
s1	0.005	-	0.13	-	-
α	90^0	105^0	90^0	105^0	-

Small Outline SOP-8



SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	0.188	0.197	4.80	5.00	-
B	0.149	0.158	3.80	4.00	-
C	0.228	0.244	5.80	6.20	-
D	0.050 BSC		1.27 BSC		-
E	0.013	0.020	0.33	0.51	-
F	0.004	0.010	0.10	0.25	-
H	0.053	0.069	1.35	1.75	-
J	0.011	0.019	0.28	0.48	
K	0.007	0.010	0.19	0.25	-
M	0.016	0.050	0.40	1.27	
L	0.150 REF		3.81 REF		-
e1	45^0		45^0		-
α	0^0	8^0	0^0	8^0	-

*All specs and applications shown above subject to change without prior notic.